

- Q.31 Discuss the calculation of cleaning efficiency of blow room and carding machines.
- Q.32 Describe the procedure to determine yarn count using the simple weighing method.
- Q.33 Explain the working and application of Beesley's yarn balance in count determination.
- Q.34 Discuss the step-by-step procedure for the Differential Dyeing Method to determine fibre maturity.
- Q.35 Differentiate between single yarn twist and ply yarn twist, giving examples of their applications in textiles.

SECTION-D

Time : 3Hrs.

SECTION-A

- Q.1 The main aim of textile testing is to ensure _____ of textile products.
- a) Cost b) Quality
c) Colour d) Packaging
- Q.2 Random sampling ensures
- a) Equal chance for all units
b) Biased results
c) Shorter fibres only
d) Testing errors
- Q.3 The Digital Fibrograph measures
- a) Length distribution b) fineness
c) Strength d) Maturity
- Q.4 _____ is small fraction which is selected to represent a population.
- a) Yarn b) Fabric
c) Sample d) none of them
- Q.5 The caustic Soda Method is used to test
- a) Maturity b) fineness
c) strength d) length distribution

- Q.36 Compare different methods of twist measurement in yarns and explain the advantages and limitations of each method.
- Q.37 Discuss the concept of the universal system of yarn numbering and its advantages in international trade. Also explain Cotton Count (Ne), Worsted Count (NeW), and Metric Count (Nm) in the indirect system with examples.
- Q.38 Discuss the relationship between fibre maturity and fibre physical properties such as strength, dyeability, and handle, Also explain the importance of determining fibre maturity in textile manufacturing and quality control.

- Q.6 Cleaning efficiency is calculated for
- Spinning machines only
 - Blow room & carding
 - Weaving looms
 - Finishing machines
- Q.7 Twist factor is calculated using
- Twist per inch & yarn count
 - Fibre fineness & length
 - Moisture regain & count
 - Yarn strength & count
- Q.8 Twist is _____ turns given to fibres bundle.
- spiral
 - round
 - smooth
 - square
- Q.9 Count strength product (CSP) is
- Yarn count x Lea strength
 - Fibre fineness x length
 - Moisture regain x tensile strength
 - Yarn count x twist
- Q.10 USTER Tensojet is used to measure single yarn _____
- twist
 - quality
 - irregularities
 - strength

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 State one objectives of quality control in textiles.
- Q.12 Why is textile testing important for customer satisfaction?
- Q.13 What is Random sampling?
- Q.14 What is biased sampling?
- Q.15 Name two types of twist.

- Q.16 Name the three main parts of a cotton fibre cross section.
- Q.17 Define maturity ratio.
- Q.18 What is the use of a Vibroscope?
- Q.19 What is twist factor?
- Q.20 Name one instrument used for trash analysis.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Discuss the role of textile testing in the research and development of textile materials.
- Q.22 Discuss the principles of random sampling and why it is essential in textile quality evaluation.
- Q.23 Compare the advantages and disadvantages of squaring, cut-squaring, and zoning techniques.
- Q.24 Discuss the principle and construction of the sorter diagram method for fibre length measurement.
- Q.25 Describe the significance of mean length, upper-half mean length, and short fibre percentage.
- Q.26 Define fibre maturity, maturity percentage, and maturity ratio. Also explain effects of immature fibres on yarn and fabric properties.
- Q.27 Explain the principle of air flow methods of determining fibre fineness.
- Q.28 Define moisture content and moisture regain, explaining their differences with formulas.
- Q.29 Explain the oven-drying method for determining moisture content in textiles.
- Q.30 Describe the procedure for trash analysis in cotton using a shirley Analyser.